



Emergency Response *optimizer* RESQNOW

A smart platform designed to save lives through rapid accident detection, real-time location sharing, and seamless coordination between emergency services.

SAVE TIME SAVE LIVES

Semicolon Troops

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The Problem statement

In India, delayed emergency response is a leading cause of preventable deaths. Critical gaps in detection, routing, and coordination cost lives every day.

Slow Accident Detection

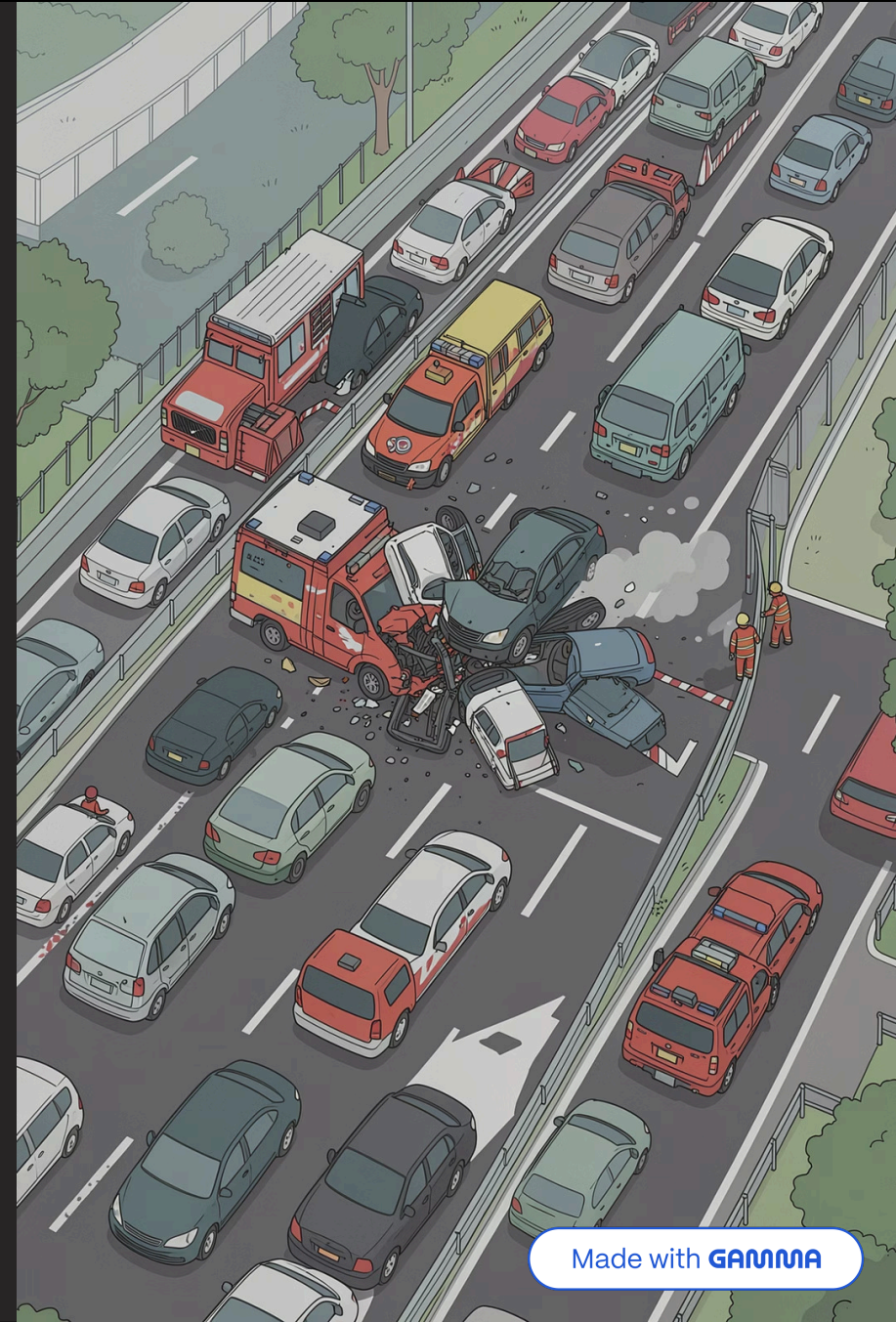
No automated system to instantly detect accidents and trigger alerts without human intervention.

No Real-Time Routing

Ambulances lack dynamic routing based on live traffic, weather, and road closures.

Bed Availability Gap

No real-time visibility into hospital bed availability leads to critical delays in patient admission.



Technical Approach

Our platform is built on a modern, scalable stack designed for speed, reliability, and real-time performance.

Mobile App (React Native)

Accident detection via speed and accelerometer sensors; live GPS tracking using Google Maps.

Backend (Node.js & Express)

Handles SOS alerts, stores data in MongoDB, and identifies the nearest hospital and ambulance.

Automation (N8N)

Sends SMS and WhatsApp alerts, notifies emergency contacts and hospitals, triggers reverse emergency workflows.

Real-Time Tracking (WebSocket)

Simultaneous live tracking of patient, doctor, and ambulance via persistent WebSocket connections.

Impacts & Benefits

Faster Response Time

Dramatically reduces the time from accident to emergency dispatch through automated detection.

Improved Survival Rate

Quicker intervention directly improves accident survival outcomes for victims.

Accurate Location Tracking

Real-time, precise GPS tracking of victims and responders ensures no time is wasted.

Better Coordination

Seamless communication between ambulances, hospitals, and emergency responders.

Traffic Optimisation

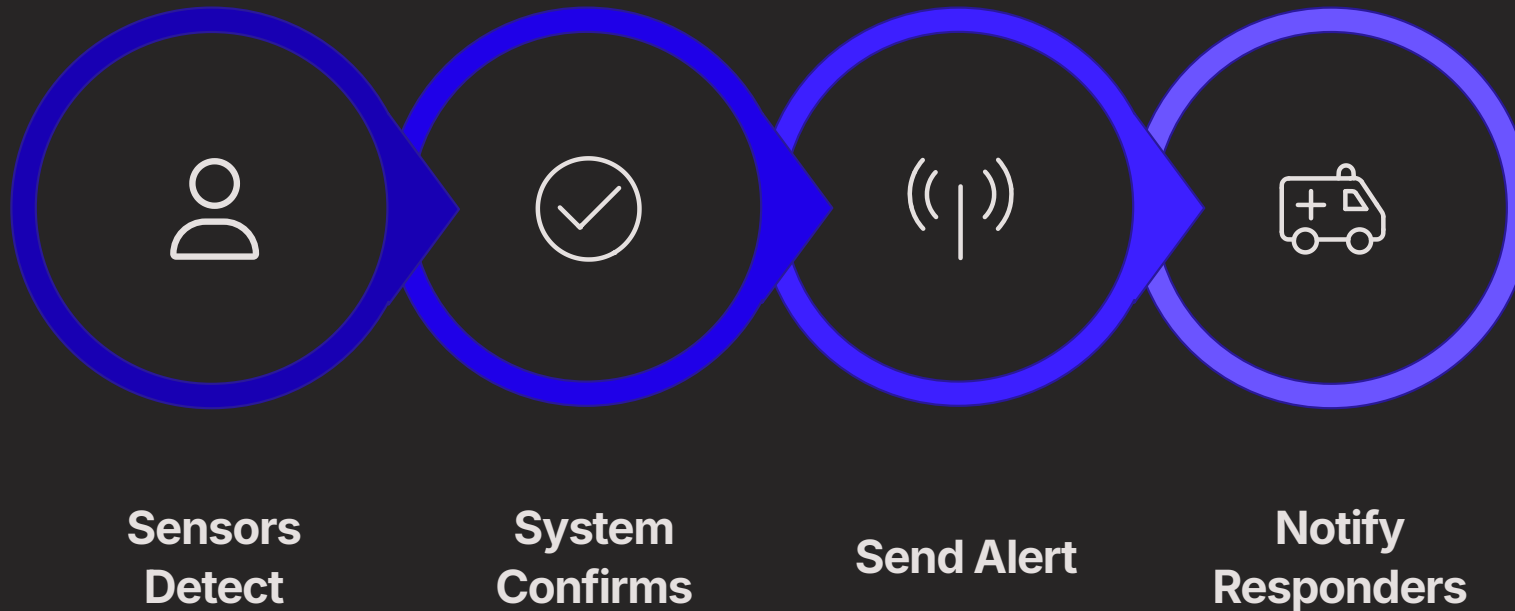
Fastest routes calculated using real-time traffic data to avoid delays.

Instant Contact Alert

Emergency contacts notified automatically and instantly via SMS and WhatsApp.

System Working Flow

From accident detection to emergency dispatch — our system operates in a seamless, automated loop.



The automated pipeline ensures zero manual delay — from the moment of impact to the dispatch of emergency services, every step is handled intelligently and instantly.

Vision & Future Scope

Our Vision

Reduce accident-related deaths through **quick, intelligent action** — connecting people, ambulances, and hospitals instantly to build a safer, more responsive transportation system across India.

Future Roadmap

- **AI Integration** — Predictive models for accident prevention and risk zones
- **Smart Device Support** — Native apps for smartwatches and smartphones
- **Live Bed Availability** — Real-time hospital capacity tracking nationwide
- **Offline Resilience** — Reduced local dependency in low-connectivity situations



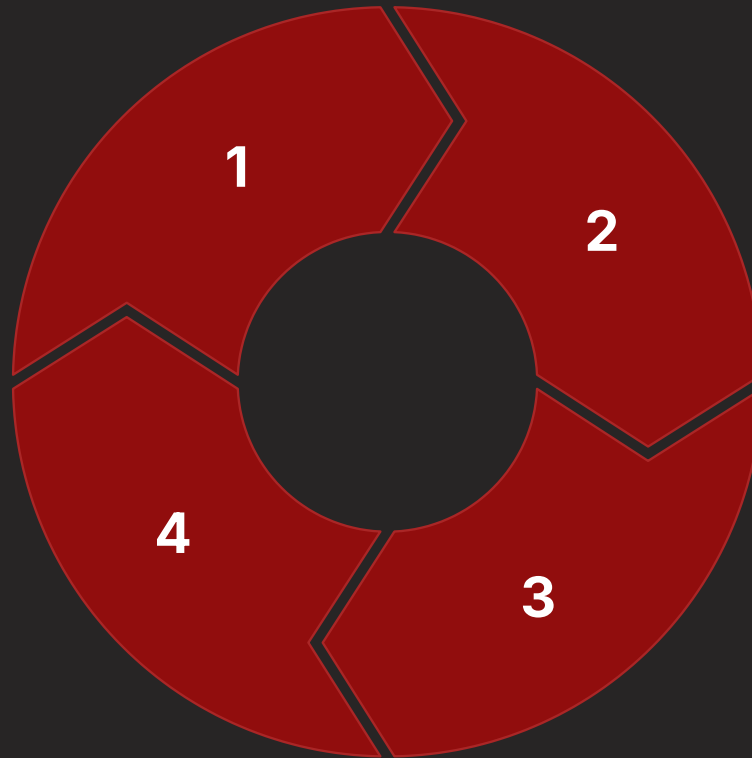
Key Takeaways

Automated Detection Saves Lives

Sensor-based accident detection eliminates human delay in raising alerts.

A Platform Built for India

Designed to address India's unique emergency response challenges with AI-powered future scope.



Real-Time Coordination is Critical

Live tracking of patients, ambulances, and hospitals ensures no gap in the chain of care.

Scalable Technology Stack

React Native, Node.js, MongoDB, N8N, and WebSocket form a robust, production-ready architecture.

THANK YOU